FinTrace

Explainable Transaction and Network Investigation Using IBM’s Synthetic AML Benchmark

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Introduction and Motivation

Title Slide

FinTrace in one line

- An explainable investigation project built on IBM’s **fully synthetic** AML benchmark.

How it finds reviewable patterns

- Uses **deterministic rules** and temporal-network evidence, not machine learning.

What makes results auditable

- Every finding links back to specific transactions and versioned policy weights.

Project status today

- Data preparation, detection, investigation, cases, local reports and historical replay are implemented.



Hidden Patterns Matter



Why single transactions mislead

- Collection can look normal until many senders appear together.
- Forwarding often matters only when receipt and send happen close.
- Dispersal hides risk when one sender feeds many recipients.



What reviewers must connect

- Time order is essential for evidence, not just connectivity.
- Counterparties reveal structure: who pays, who receives, and when.
- Multi-hop movement needs linked transfers across accounts and time.



How to interpret patterns

- A pattern is a **reason for review**, not a conclusion.
- Context can explain behavior: payroll, bill pay, or pooling activity.
- FinTrace prioritizes inspectable evidence over hidden assumptions.

What FinTrace Aims For



Outcome FinTrace targets

- Turn validated transactions into traceable findings that support consistent review.
- Keep evidence tied to specific transactions, accounts, and time ordering.

Integrity first, always

- Prioritize auditing, provenance, validation, and normalization before detection.
- Treat data quality issues as visible diagnostics, not hidden fixes.

Explainability by design

- Make each finding readable as rules plus supporting transactions.
- Show how fixed policy weights add up to a review priority.

Implemented local workflow

- Rules, graphs, priorities, alerts, cases, reports, replay and seven UI pages.

Current Status and Data Overview

Current Project Status



Implemented data foundation

- Source files are audited with SHA-256 checks and provenance manifests.
- Transactions are validated, normalized to Parquet, and tracked in an account registry.
- Benchmark labels are stored separately from label-free transaction artifacts.

Implemented import diagnostics

- Publisher-account metadata is preserved for reconciliation and link diagnostics.
- Completed imports are reused only after checksum-verifying all artifacts.

Implemented investigation services

- Six transaction rules, temporal graphs and fixed-policy priorities.
- Audited alerts, cases, local reports and checkpointed replay.

Current UI reality

- Seven pages; HI-Medium snapshot: 280 findings, 269 scored accounts, 270 alerts.

Snapshot at a Glance

Verified HI-Medium dataset

31,898,238

Normalized transactions

Notes: Accepted 1:1 from source

2,077,023

Observed account keys

Notes: Identity: bank_id + account_id

15

Payment currencies

Notes: No cross-currency totals

7

Payment formats

Notes: Format preserved from source

2022/09/01 00:00 to 2022/09/28 15:58

Timestamp bounds (min precision)

Notes: Bounds do not prove full coverage

Labels Stay Separate

Purpose of the benchmark labels

- Source labels remain isolated from detection, scoring, investigation, reports and replay.
- Benchmark evaluation is outside the maintained scope.

HI-Medium label validation

- All 31,898,238 source labels are valid binary values.
- No label issues were recorded; artifacts stay separate.

How to interpret label 1

- These are synthetic laundering annotations, not confirmed mule accounts.
- They indicate patterns to test against, not operational investigation outcomes.



Data Preparation and Integrity

How Data Is Prepared

Trustworthy inputs

- Source files stay immutable, with audit logs and SHA-256 hashes.
- Versioned provenance manifests capture what was imported and when.

Validated normalization

- Headers are checked before financial fields are accepted for processing.
- Validated transactions are converted from CSV to Parquet in chunks.

Separated outputs

- Label-free transactions and observed accounts are produced for analysis.
- Diagnostics, quarantine lane, labels and publisher metadata remain isolated.

Safe reuse on re-import

- Completed imports are reused only after checksum verification succeeds.

Integrity Before Analysis

01

Snapshot validation result

- 31,898,238 source rows matched 31,898,238 accepted transactions.
- 0 financial rows quarantined in the HI-Medium dataset.

02

Integrity-preserving normalization

- Amounts remain exact using fixed-precision decimal representation throughout processing.
- Account identity is preserved as (bank_id, account_id) strings, unchanged.

03

Diagnostics kept, not judged

- All 20 duplicate-content rows are retained, not removed.
- All 2,561,860 self-transfers are retained without asserting suspicion.

Architecture and Implemented Features

Architecture in Layers

Implemented data foundation

- Ingestion, validation and local artifacts are implemented and checksum-verifiable.
- Benchmark labels and publisher metadata remain physically separated from analysis data.

Implemented investigation services

- Rules and temporal graphs generate evidence-backed findings.
- Priorities, audited alerts, cases, reports and replay support review.

Clear separation of concerns

- Domain services compute findings; Streamlit presents results and submits explicit analyst actions.
- Labels remain isolated; benchmark metrics are outside scope.

Core technology stack

- Python 3.11+, PyArrow/Parquet, DuckDB, Pandas, NetworkX, Streamlit and TOML policies.

Six Deterministic Transaction Rules

Rule	Trigger (illustrative)	Primary evidence	Coverage constraint
Pass-through	3 inbound; 80% forwarded; 60 min	FIFO inflow to outflow allocation	Requires strict timestamp order
Fan-in / Fan-out	8 senders or 8 recipients; 60 min	Distinct counterparties per window	Currency handled separately
Burst vs history	20 pays; 4x median	Count vs historical median	Needs 7 prior covered days
Inactivity activation	7 days quiet; then 10 pays; 120 min	Gap then concentrated activity	Depends on coverage during gap
Repeated splitting	5 payments same directed pair; 60 min	Repeated A to B transfers	Does not prove evasion intent

Demonstration thresholds, not legal requirements. Burst and dormant activation are unavailable in this snapshot: historical coverage is unknown.

Pass-Through Example

Illustrative event sequence

- Three inflows of 100 units arrive before any forwarding occurs.
- One later outflow forwards 250 units within 60 minutes.

FIFO allocation result

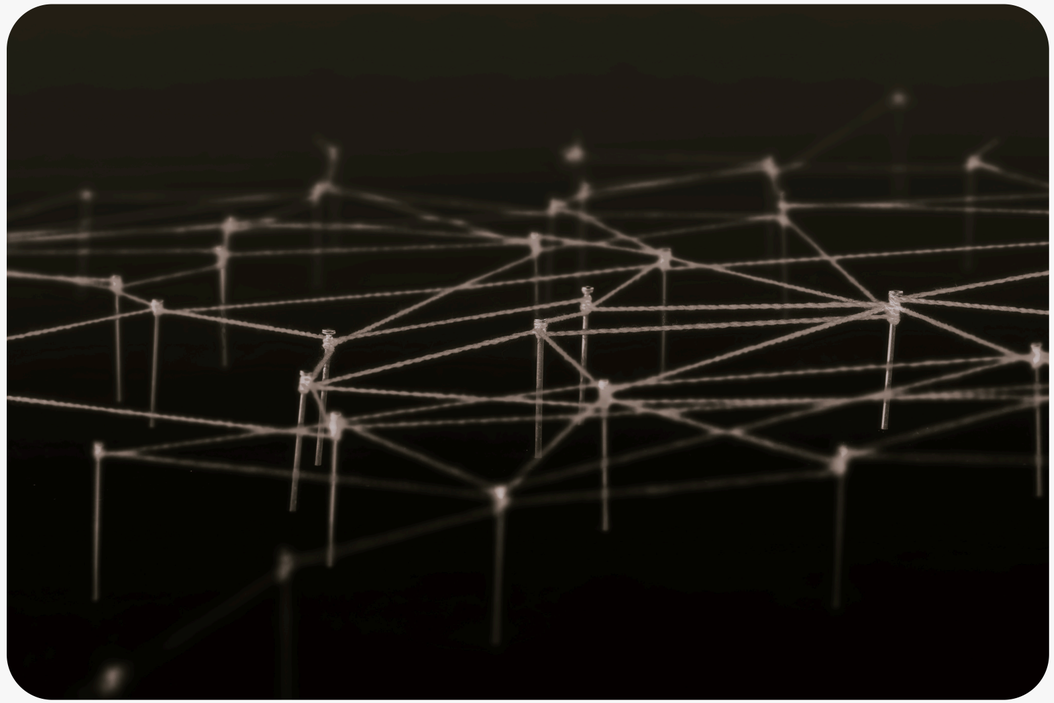
- Eligible observed inflow totals 300; allocated forwarding totals 250.
- Forwarded share is **83.33%** (250 divided by 300).

Interpretation limits

- This finding requires strict timestamp order; reversing it removes evidence.
- Allocation supports review, not exact source-of-funds proof.



Temporal Graphs Implemented



Represent transactions as a temporal network

- Model each transfer as a directed edge with ID, timestamp, amount, and currency.
- Keep parallel edges so repeated transfers remain distinct evidence items.

Use graph methods to narrow reviewable structures

- Find chronological chains, cycles and repeated coordinated groups using bounded searches and explicit conditions.
- Retain ordered transaction IDs and disclose truncated coverage.

Treat topology as context, not proof

- Communities and centrality provide descriptive context for where to look next.
- Connectivity alone does not establish suspicious behavior or a money-flow chain.

Time Order Is Crucial



Require strict time order

- Treat linked transfers as evidence only when timestamps increase step by step.
- If the order is not provable, report connectivity without claiming a path.

Handle same-minute ties safely

- Minute-level timestamps mean equal times cannot establish who paid first.
- Group tied events and avoid attributing within-minute forwarding sequences.

Cycles need chronological support

- A cycle is reviewable only when each hop occurs later than the last.
- Graph structure alone does not show movement direction over time.

Disclose bounds and truncation

- Cap search depth, path length, and expansion to keep runtime predictable.
- If search stops early, label results as partial with stated limits.

Priority Must Stay Explainable

What the 0–100 priority means

- A fixed-policy review rank, not a prediction or likelihood.
- Computed from explainable evidence families, each with set weights.

Illustrative score build-up

- Pass-through finding contributes **+25** under the demo policy.
- Collection or dispersal family contributes **+15** when triggered.
- Temporal-network evidence contributes **+30** when supported by time order.

Example total and guardrails

- Total equals **70/100**, using min(100, sum of eligible contributions).
- At most one contribution per family; overlaps do not double count.
- High priority means “review sooner,” not “confirm suspicious behaviour.”



Investigation Workflow and Interface

Alerts and Cases Implemented



Implemented alert lifecycle

- Alerts move from open to in_review, then escalated or closed.
- Reopening is allowed, but must be explicit and audited.

Implemented case lifecycle

- Open cases can move to investigating or close; investigating cases can escalate or close. Reopening is explicit.
- Frozen attachments, assignment, notes and immutable versions retain supporting evidence and decision history.

Governance and traceability

- Writes require actor, time, reason and expected version; case request IDs make retries idempotent.
- Escalation is internal; reports remain local.

Seven-Page Dashboard

Filters to focus analyst review

• Select a verified snapshot; inspect dataset, cutoff and policy identity.
• Filter alerts by currency, evidence family, priority band and status.

One workspace, multiple evidence views

• Overview, Alert review, Account evidence and Network explorer.
• Cases, Reports and Replay complete the seven-page workspace.
• Account transfers cover the selected graph window only.

Coverage and ordering are always visible

• Top ten active-alert accounts seed the combined network view.
• Defaults: six display hops, up to 100 accounts and 150 transfers.
• Fixed layout, zoom, full screen and explicit bounded-view notices.

Labels never appear in investigations

• Benchmark labels and pattern annotations stay outside investigations.



Traceability and Exports



End-to-end traceability

- Each review item links source record, finding, alert, and case.
- Every score contribution points to specific supporting transactions.



Export package contents

- Include dataset/run identity, source hashes, analysis cutoff and applied policy.
- Publish atomically with a checksummed manifest.



Readable evidence outputs

- Local JSON, CSV, HTML, PNG; verified ZIP downloads.
- Escape HTML; neutralize CSV formulas; verify evidence.
- No regulatory submission.

Context, Replay, and Scope

What Context Exists

What we can observe from transactions

- Each record provides sender and recipient accounts and bank IDs.
- We retain timestamp, amount, currency, and payment format fields.

Supplemental synthetic context (separate lane)

- Publisher metadata is preserved for reconciliation diagnostics.
- Link candidates do not rewrite keys or verify customer identity.

What we do not have (without external sources)

- No verified customer identity, ownership, devices, IPs, or geography.
- Independent context and screening are outside maintained scope.



Replay and Verification

Historical replay (implemented)

- Reuse bounded rule/graph services at exclusive historical cutoffs.
- Tied timestamps cannot establish a transfer sequence.
- Isolate replay alerts; verify source/policy identity on resume.

Freeze evidence over time

- Once emitted, earlier findings stay unchanged as later data arrives.
- Updates append new evidence; they do not rewrite prior outputs.

Verified replay and remaining scope

- HI-Medium replay matched batch: 280 findings, 269 scored accounts and 270 alerts at the exclusive cutoff 2022-09-03T06:00:00Z.
- Full ingestion is not full-history detection; coverage limits remain.
- Benchmark accuracy metrics are outside the maintained scope.



Implemented Services and Scope

1. Data foundation Implemented

Validated, versioned artifacts with audit and provenance

2. Explicit transaction rules Implemented

Codify explicit patterns into reproducible findings

3. Temporal graph evidence Implemented

Chronological paths and cycle candidates with bounds

4. Priorities, alerts and cases Implemented

Policy-weighted review ranking and audited workflows

5. Dashboard and exports Implemented

Service-driven UI plus CSV/JSON/HTML /PNG artifacts

6. Historical replay Implemented

Bounded replay; verified resume. Metrics and independent context remain outside scope.